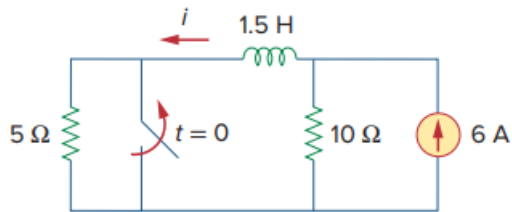


Problem

The switch in Fig. 7.52 has been closed for a long time. It opens at $t = 0$. Find $i(t)$ for $t > 0$.

Fig. 7.52:



Step-by-step solution

Step 1 of 8

Consider the following circuit diagram:

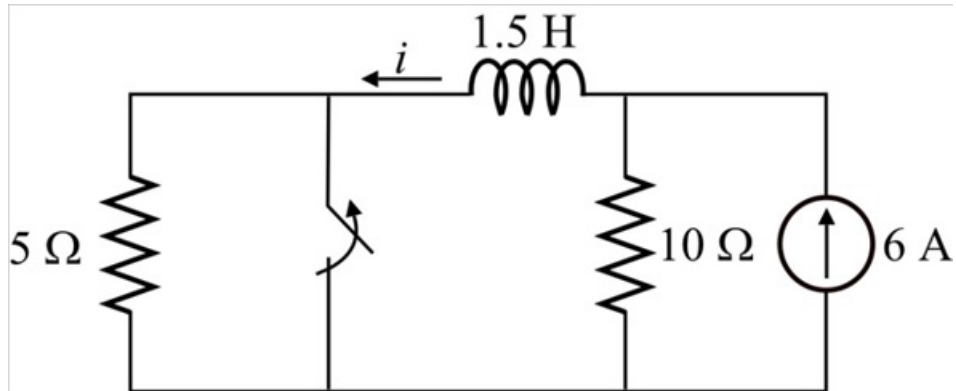


Figure 1

Step 2 of 8

For $t = 0^-$, the switch in the circuit is closed and inductor acts like short circuit. Draw the modified circuit diagram:

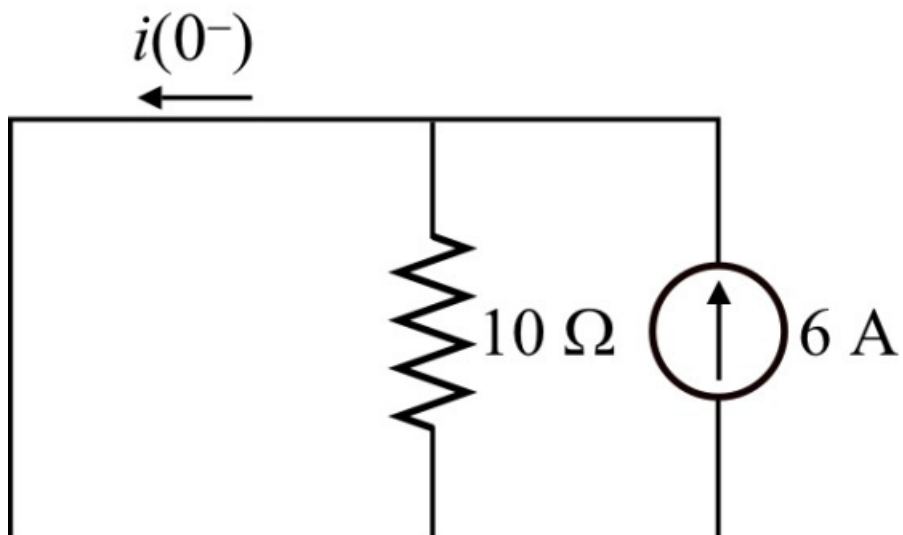


Figure 2

Step 3 of 8

The total 6 A source is directly flows through short circuit only. Therefore, the current flows through short circuit is 6 A.

$$i(0^-) = 6 \text{ A}$$

Step 4 of 8

Since the inductor current cannot change instantaneously. $i(0^-) + i(0^+) = i(0) = 6 \text{ A}$

Step 5 of 8

For $t > 0$, the switch in the circuit is open and inductor acts like short circuit. Draw the modified circuit diagram.

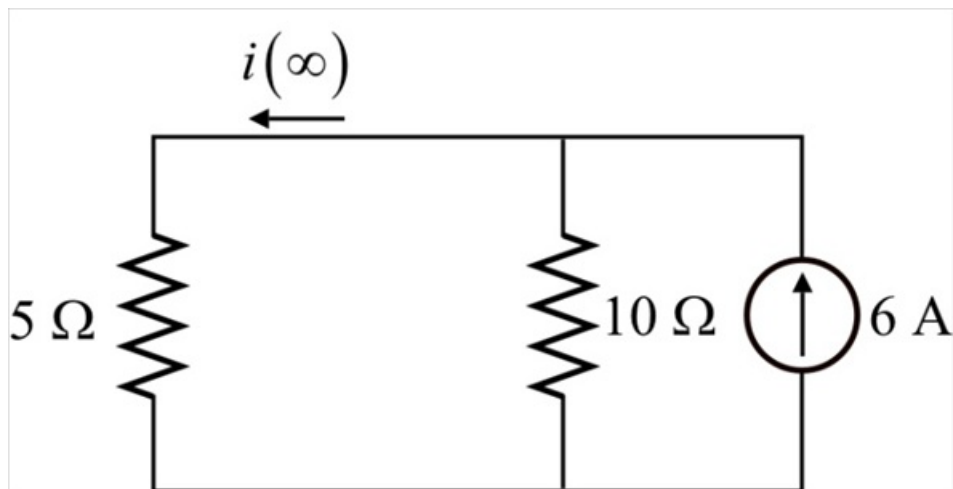


Figure 3

Step 6 of 8

Calculate the current through inductor for $t = \infty$.

$$\begin{aligned} i(\infty) &= \left(\frac{10}{5+10} \right) (6) \\ &= \left(\frac{10}{15} \right) (6) \\ &= (0.667)(6) \\ &= 4 \text{ A} \end{aligned}$$

Step 7 of 8

To calculate the equivalent resistance at $t > 0$, open the current source and draw the modified circuit.

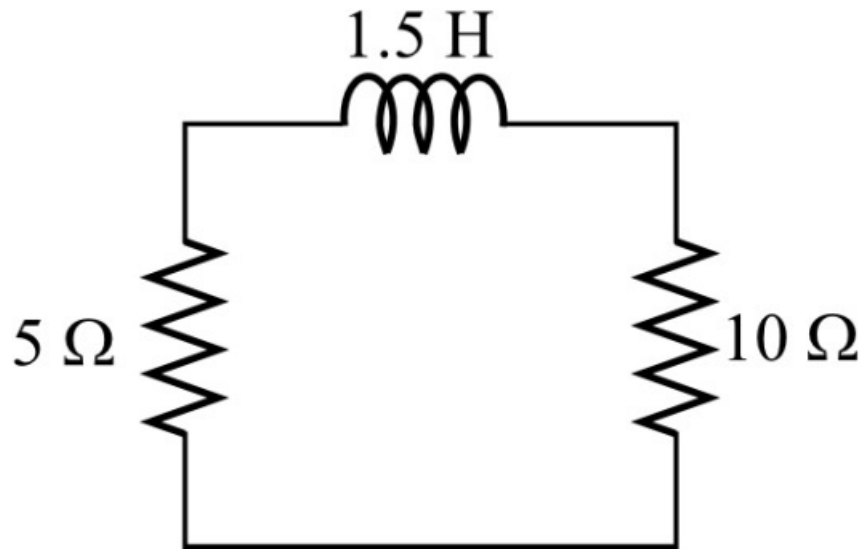


Figure 4

Step 8 of 8

Calculate the equivalent resistance across the inductor terminals from Figure 4.

$$\begin{aligned} R_{th} &= 5 + 10 \\ &= 15 \, \Omega \end{aligned}$$

Calculate time constant.

$$\begin{aligned} \tau &= \frac{L}{R_{th}} \\ &= \frac{1.5}{15} \\ &= \frac{1}{10} \, \text{s} \end{aligned}$$

Calculate the current through the inductor for $t > 0$.

$$\begin{aligned} i(t) &= i(\infty) + (i(0) - i(\infty))e^{-\frac{t}{\tau}} \\ &= 4 + (6 - 4)e^{-\left(\frac{1}{10}\right)t} \\ &= 4 + 2e^{-10t} \, \text{A} \end{aligned}$$

Therefore, the current through inductor, $i(t)$ for $t > 0$ is $\boxed{(4 + 2e^{-10t}) \, \text{A}}$.